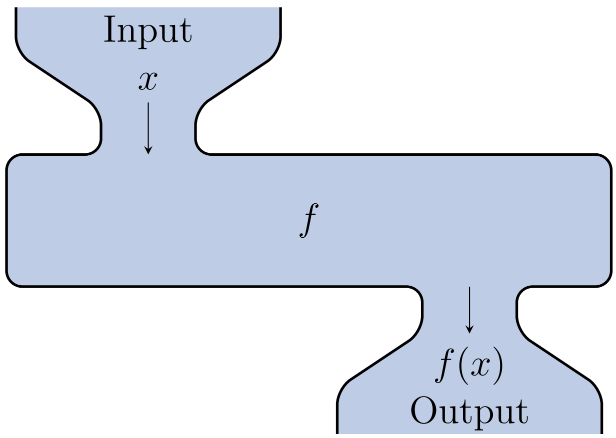
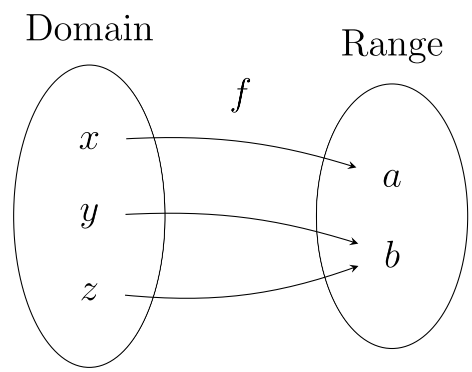
1.1: Functions and Function Notation

Definition.
A **function** is a relation in which each possible input value leads to exactly one output value.
The input values make up the **domain**, and the output values make up the **range**.





Example. Let $f(x) = 2x^2 - 2x + 1$. Evaluate the following

$$f(1) = 2(1)^2 - 2(1) + 1 = 2(1) - 2 + 1 = 2(1) - 1 = 1 \rightarrow \boxed{f(1)=1}$$

$$f(-2) = 2(2)^2 - 2(2) + 1 = 2(4) - 4 + 1 = 8 - 3 = 5 \rightarrow \boxed{f(-2)=5}$$

$$f(a) = 2(a)^2 - 2(a) + 1 = \boxed{2a^2 - 2a + 1}$$

$$f(a+h) = 2(a+h)^2 - 2(a+h) + 1 = 2(a+h)(a+h) - 2a - 2h + 1 = 2(a^2 + 2ah + h^2) - 2a - 2h + 1 = \boxed{2a^2 + 4ah + 2h^2 - 2a - 2h + 1}$$

$$\frac{f(a+h) - f(a)}{h} = \frac{[2a^2 + 4ah + 2h^2 - 2a - 2h + 1] - [2a^2 - 2a + 1]}{h} = \frac{4ah + 2h^2 - 2h}{h} = \boxed{-4a - 2h + 2}$$

Example. Suppose that the following tables represent a carnival truck's menu prices. Which table would represent a valid function?

Item	Price
Funnel cake	\$8
Chicken strips	\$9
Ribbon fries	\$7
Cheese nachos	\$6

Valid function!

Item	Price
Funnel cake	\$8
Chicken strips	\$8
Ribbon fries	\$7
Cheese nachos	\$6

Valid function!
(repeats prices are ok)

Item	Price
Funnel cake	\$8
Chicken strips	\$8
Ribbon fries	\$8
Funnel cake	\$6

Not a valid function

Example. Based on an old match-makers rule, the socially "acceptable" minimum age of someone you can date is half your age plus seven. Make a function that takes your age, x, and finds the minimum age of someone you can date, y.

$$y = f(x) = \frac{x}{2} + 7$$

your age

prospective partner's age

Q: Assuming both partners follow this rule, what's the maximum age you could date?

Example. Create a table to represent a function that takes each month and returns the number of days (in a non-leap year).

Month	1	2	3	4	5	6	7	8	9	10	11	12
Days	31	28	31	30	31	30	31	31	30	31	30	31

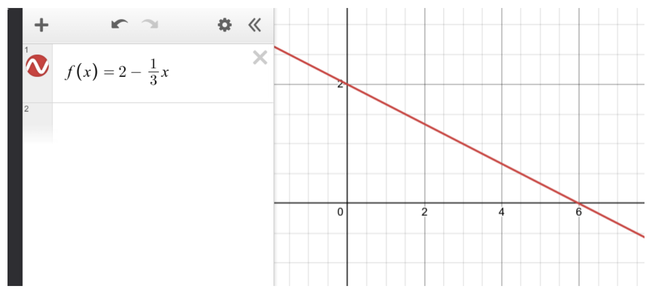
e.g. $f(6) = 30$

Example. If possible, express the relationship $2n + 6p = 12$ as a function $p = f(n)$. Graph the result.

Solve for p:

$$-2n + 2n + 6p = 12 - 2n$$

$$\frac{6p}{6} = \frac{12-2n}{6}$$



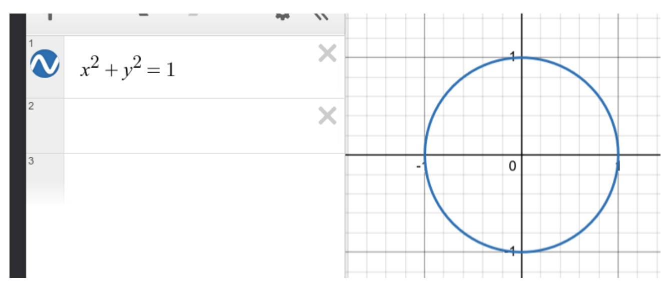
$$p = \frac{12-2n}{6} = \frac{12}{6} - \frac{2n}{6} = 2 - \frac{1}{3}n \longrightarrow \boxed{f(n) = 2 - \frac{1}{3}n}$$

Example. If possible, express the relationship $x^2 + y^2 = 1$ as a function $y = f(x)$. Graph the result.

$$x^2 + y^2 = 1$$

$$y^2 = 1 - x^2$$

$$y = \pm\sqrt{1-x^2}$$

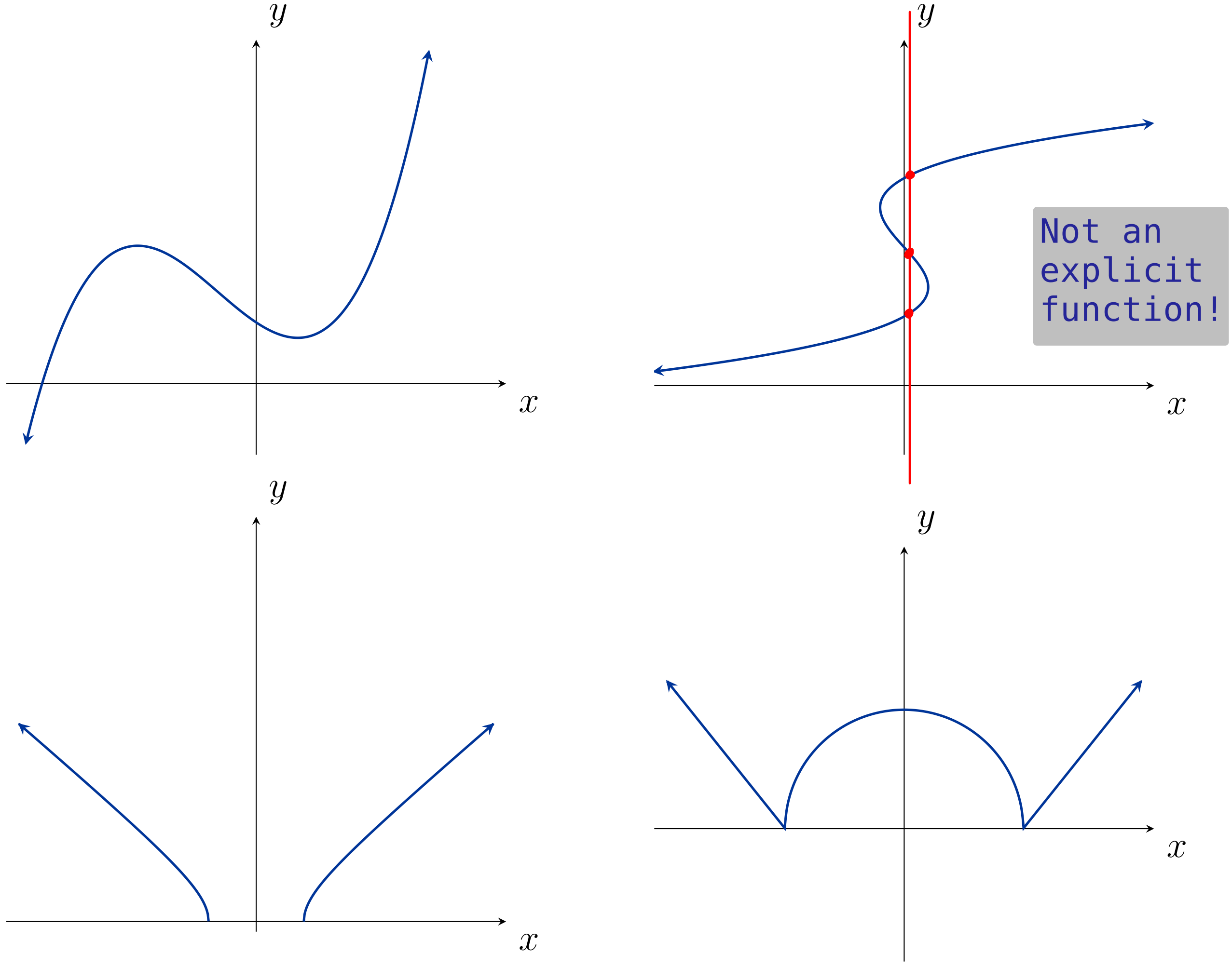


$\boxed{y = \sqrt{1-x^2}}$ $\boxed{y = -\sqrt{1-x^2}}$

Not an EXPLICIT function of x

Definition. (Vertical Line Test)
A curve in the xy-plane is the graph of a function $y = f(x)$ (an explicit function) if and only if each vertical line intersects it in at most one point

Example. Use the vertical line test on the following graphs to determine which graphs may represent an explicit function:



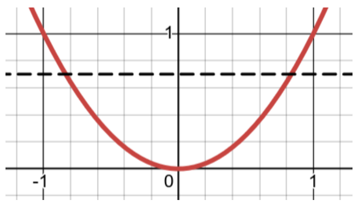
Definition. (One-to-One)
The **horizontal line test** determines if every horizontal line passes through the graph of a function in at most one point.
A function is said to be **one-to-one** if each output value corresponds to *exactly* one input value (and hence passes the horizontal line test).

Example. Which of the following represent a one-to-one function? Graph

$y = x^2$

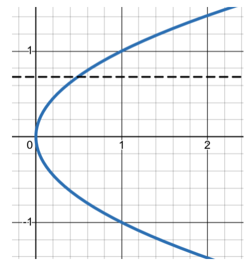
Valid Function

Not one-to-one



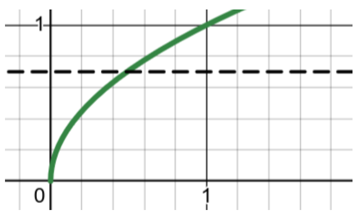
$y^2 = x$

Not an explicit function



$f(x) = \sqrt{x}$

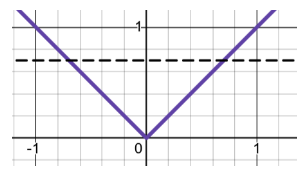
One-to-one



$y = |x|$

Valid Function

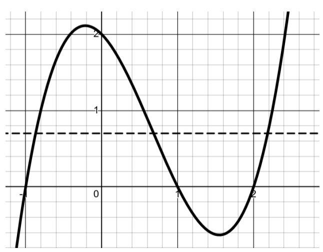
Not one-to-one



$f(x) = (x+1)(x-1)(x-2)$

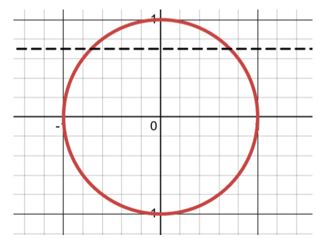
Valid Function

Not one-to-one



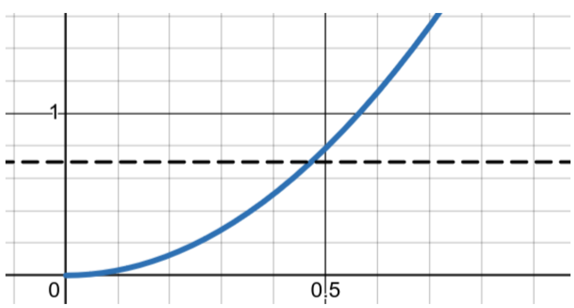
The equation of a circle:
$x^2 + y^2 = 1$

Not an explicit function



The area of a circle:
$A = \pi r^2$

One-to-one



Graphs of basic functions